

SECTION 08 11 16

ALUMINUM DOORS AND FRAMES
05/17, CHG 2: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings
Except Low-Rise Residential Buildings

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon
Structural Steel

ASTM B209 (2014) Standard Specification for Aluminum
and Aluminum-Alloy Sheet and Plate

ASTM B209M (2014) Standard Specification for Aluminum
and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221 (2021) Standard Specification for Aluminum
and Aluminum-Alloy Extruded Bars, Rods,
Wire, Profiles, and Tubes

ASTM B221M (2021) Standard Specification for Aluminum
and Aluminum-Alloy Extruded Bars, Rods,
Wire, Profiles, and Tubes (Metric)

ASTM E283 (2019) Standard Test Method for
Determining the Rate of Air Leakage
Through Exterior Windows, Curtain Walls,
and Doors Under Specified Pressure
Differences Across the Specimen

ASTM E331 (2000; R 2023) Standard Test Method for
Water Penetration of Exterior Windows,
Skylights, Doors, and Curtain Walls by
Uniform Static Air Pressure Difference

ASTM E1886 (2019) Standard Test Method for
Performance of Exterior Windows, Curtain
Walls, Doors, and Impact Protective
Systems Impacted by Missile(s) and Exposed
to Cyclic Pressure Differentials

ASTM E1996 (2017) Standard Specification for
Performance of Exterior Windows, Curtain
Walls, Doors, and Impact Protective

Systems Impacted by Windborne Debris in
Hurricanes

NATIONAL FENESTRATION RATING COUNCIL (NFRC)

NFRC 100

(2020) Procedure for Determining
Fenestration Product U-Factors

NFRC 200

(2020) Procedure for Determining
Fenestration Product Solar Heat Gain
Coefficient and Visible Transmittance at
Normal Incidence

1.2 PERFORMANCE REQUIREMENTS

1.2.1 Structural Calculations

1.2.2 Wind Borne Debris

Provide impact resistant door assemblies meeting the
Windborne-Debris-Impact Resistant Performance requirements of
ASTM E1996 and ASTM E1886 as follows:

- (1) Pass missile-impact tests when tested according to ASTM E1886 and
meeting performance requirements according to ASTM E1996 for
missiles A and D in Table 2.

1.2.3 Air Infiltration

When tested in accordance with ASTM E283, air infiltration per door leaf
cannot exceed 0.6 cubic feet per minute per square foot of fixed area at a
test pressure of 6.24 pounds per square foot.

1.2.4 Water Penetration

When tested in accordance with ASTM E331, there can be no water
penetration at a pressure of 2.86 pounds per square foot of fixed area.

1.2.5 Thermal Transmittance, Solar Heat Gain, Visible Light Transmittance

Provide products bearing NFRC Project Label Certificates for Fenestration
verifying compliance with requirements for each assembly indicated. An
NFRC Bid Report, or approved equal, for field assembled exterior doors may
be submitted in lieu of Project Label Certificates for Fenestration if
such reports are created in accordance with NFRC CAMP procedures and are
provided by the manufacturer. Such alternate reports may be submitted
with shop drawings, however, NFRC validated Project Label Certificates for
Fenestration are required as a Closeout Submittal. Contact NFRC for
information on NFRC 100 and NFRC 200 Compliance and Monitoring Program
(CAMP) rating requirements:

<http://www.nfrc.org/industry/certification/compliance-and-monitoring-program-camp/>

1.2.5.1 U-Factor

Provide exterior glazed assemblies, including aluminum entrances doors
with greater than 50 percent glazed area, certified by the NFRC as having
a whole window U-factor of 0.57 or less as determined in accordance with
ASHRAE 90.1 - IP and as verified in accordance with NFRC 100.

1.2.5.2 Solar Heat Gain Coefficient (SHGC)

Provide exterior glazed assemblies, including aluminum entrances doors with greater than 50 percent glazed area, certified by the National Fenestration Rating Council with a whole window SHGC of 0.39 or less as determined in accordance with ASHRAE 90.1 - IP and as verified in accordance with NFRC 200.

1.2.5.3 Visible Light Transmittance (VLT)

Provide exterior glazed assemblies, including aluminum entrances doors with greater than 50 percent glazed area, certified by the NFRC with a whole window VLT of 43% or greater as determined in accordance with ASHRAE 90.1 - IP and as verified in accordance with NFRC 200.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

For Each Type of Door and Frame Assembly; G

Delegated Design; G

SD-03 Product Data

For Each Type of Door and Frame Assembly; G

Recycled Content of Aluminum Material; S

SD-04 Samples

Finish Samples; G

SD-05 Design Data

Structural Calculations for Deflection and Antiterrorism; G

SD-06 Test Reports

Air Infiltration; G

Water Penetration; G

SD-07 Certificates

NFRC Project Label Certificates for Fenestration; G

SD-08 Manufacturer's Instructions

Installation of Each Type of Door and Frame Assembly; G

SD-10 Operation and Maintenance Data

Adjustments, Cleaning, and Maintenance; G

SD-11 Closeout Submittals

NFRC Project Label Certificates for Fenestration; G

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect materials delivered to the site for damage. Unload and store with minimum handling. Provide storage space in dry location with adequate ventilation, free from dust or water, and easily accessible for inspection and handling. Stack materials on non-absorptive strips or wood platforms. Do not cover doors and frames with tarps, polyethylene film, or similar coverings. Protect finished surfaces during shipping and handling using manufacturer's standard method. Do not apply coatings or lacquers to surfaces to which caulking and glazing compounds must adhere.

1.5 QUALITY CONTROL

1.5.1 Delegated Design

Submit a DELEGATED DESIGN demonstrating compliance.

Furnish aluminum doors and frames that resist uplift and lateral loads. Marine Corps Air Station Cherry Point must resist a 139 mile per hour, ultimate design wind speed, 50 PSF, Risk Category II, Surface Roughness Category C, Exposure Category C lateral load. Comply with UFC 3-301-01 and ASCE 7 as applicable for building system designs and components. Assembly design responsibility is delegated to the Contractor and incorporates specific properties and structural capabilities of selected products and manufacturers.

Supplement delegated design shop drawings with storefront system structural computations. Tie general manufacturer data of appurtenant members utilized to project specific conditions. Specify fastening patterns and connections.

1.5.2 Shop Drawing

Indicate elevations and sections for each type of door and frame assembly. Show sizes and details of each assembly, frame construction, thickness and gages of metal, details of door and frame construction, proposed method(s) of anchorage, glazing details, provisions for an location of hardware, mullion details, method and materials for flashing and weatherstripping, miscellaneous trim, installation details, and other related items necessary for a complete representation of all components. A qualified blast engineer must perform testing or calculations for door system design resistance to specified loads.

1.5.3 Finish Samples

Submit two color charts and two finish sample chips for each type of finish indicated.

1.5.4 Operation and Maintenance Data

Submit detailed instructions for installation, adjustments, cleaning, and maintenance of each type of assembly indicated.

PART 2 PRODUCTS

2.1 DOORS AND FRAMES

Provide swing-type aluminum doors and frames of size, design, and location indicated. Provide doors complete with frames, framing members, trim, and accessories. Coordinate side lites, window walls, adjacent curtainwall with Section 08 41 13 ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS.

2.2 MATERIALS

2.2.1 Anchors

Stainless steel or steel with hot-dipped galvanized finish.

2.2.2 Weatherstripping

Continuous wool pile, silicone treated, or type recommended by door manufacturer.

2.2.3 Aluminum Alloy for Doors and Frames

ASTM B221M, ASTM B221, Alloy 6063-T5 for extrusions. ASTM B209M, ASTM B209, alloy and temper best suited for aluminum sheets and strips. Provide aluminum materials that include a minimum of 30 percent recycled content. Provide data indicating percentage of recycled content of aluminum material.

2.2.4 Fasteners

Hard aluminum or stainless steel.

2.2.5 Structural Steel

ASTM A36/A36M.

2.2.6 Finish

Give aluminum extrusions a pretreatment to enhance adhesion followed by a caustic etch or alkaline wash for cleaning and degreasing. Apply a phosphate spray or chromate conversion treatment to protect against humidity and corrosive chemicals. Furnish an anodized treatment conforming to AAMA 611 to obtain an architectural Class I coating. Provide a two coat application that achieves a 0.7 mil thickness.

Aluminum doors and frames are clear anodized as indicated in the Marine Corps Air Station Cherry Point, Base Exterior Architectural Plan, Section 4.3. Submit aluminum door and frame selections to the Architectural Review Board prior to making submittal.

2.3 FABRICATION

2.3.1 Aluminum Frames

Extruded aluminum shapes with contours approximately as indicated. Provide removable glass stops and glazing beads for frames accommodating fixed glass. Use countersunk stainless steel Phillips screws for exposed fastenings, and space not more than 12 inches on center. Mill joints in frame members to a hairline fit, reinforce, and secure mechanically.

2.3.2 Aluminum Doors

Of type, size, and design indicated and minimum 1-3/4 inch thick. minimum wall thickness, 0.125 inch, except beads and trim, 0.050 inch. Door sizes shown are nominal; include standard clearances as follows: 0.093 inch at hinge and lock stiles, 0.125 inch between meeting stiles, 0.125 inch at top rails, 0.187 inch between bottom and threshold, and 0.687 inch between bottom and floor. Provide bevel single-acting doors 0.063 or 0.125 inch at lock, hinge, and meeting stile edges.

2.3.2.1 Full Glazed Stile and Rail Doors

Provide doors with stiles and rails as indicated. Fabricate from extruded aluminum hollow seamless tubes or from a combination of open-shaped members interlocked or welded together. Fasten top and bottom rail together by means of welding or by 3/8 or 1/2 inch diameter cadmium-plated tensioned steel tie rods. Provide an adjustable mechanism of jack screws or other methods in the top rail to allow for minor clearance adjustments after installation.

2.3.3 Welding and Fastening

Provide door corner construction consisting of mechanical clip fastening, SIGMA deep penetration plug welds and 1 1/8 inch long fillet welds inside and outside all four corners. Provide a hook-in type exterior glazing stop with EPDM glazing gaskets reinforced with non-stretchable cord. Provide an interior glazing stop that is mechanically fastened to the door member and that incorporates a silicone-compatible spacer used with silicone sealant.

Where possible, locate welds on unexposed surfaces. Dress welds on exposed surfaces smoothly. Select welding rods, filler wire, and flux to produce a uniform texture and color in finished work. Remove flux and spatter from surfaces immediately after welding. Exposed screws or bolts will be permitted only in inconspicuous locations, and must have countersunk heads. Weld concealed reinforcements for hardware in place.

2.3.4 Weatherstripping

Equip meeting stiles on pairs of doors with an adjustable astragal using wool pile with a polymeric fin. Door weatherstripping consists of a thermoplastic elastomer weatherstripping on a tubular shape with a semi-rigid polymeric backing. Provide on stiles and rails of doors. Fit into slots which are integral with doors or frames. Provide EPDM blade gasket sweep strips in an aluminum extrusions applied to bottom rails with concealed fasteners. Weatherstripping must be replaceable without special tools, and adjustable at meeting rails of pairs of doors. During installation, verify doors swing freely and close positively. Provide weatherstripping that meets specified performance tests. Refer to paragraph AIR INFILTRATION for air leakage requirements and testing.

2.3.5 Anchors

On the backs of subframes, provide anchors of the sizes and shapes indicated for securing subframes to adjacent construction. Anchor transom bars at ends and mullions at head and sill. Reinforce and anchor freestanding door frames to floor and wall constructions. Place anchors near top and bottom of each jamb and at intermediate points not more than

25 inch apart.

2.3.6 Provisions for Hardware

Coordinate with Section 08 71 00 DOOR HARDWARE. Deliver hardware templates and hardware (except field-applied hardware) to the door manufacturer for use in fabrication of aluminum doors and frames. Cut, reinforce, drill, and tap doors and frames at the factory to receive template hardware. Provide doors to receive surface-applied hardware, except push plates, kick plates, and mop plates, with reinforcing only; drill and tap in the field. Provide hardware reinforcements of stainless steel or steel with hot-dipped galvanized finish, and secure with stainless steel screws.

Concealed leaf roton hinges are heavy duty, complying with ANSI A 156.26, A31031G for high frequency or heavy medium frequency use doors. Provide a hinge side clearance of 0.3125 inches in addition to standard lock stile clearances. Door or frame reinforcement is not required for doors weighing up to 200 pounds. Door weights in excess of 200 pounds require 16 gauge channel frame reinforcement. Hinges are UL listed for 3 Hours. Provide an extruded aluminum threshold, one piece per door opening, with ribbed surface.

Concealed retail deadlocks have aluminum faceplates and eight ply, laminated stainless steel bolts with 1.375 inch throw. Locks accept standard 1.1563 inch mortise cylinders for keyed exterior access and thumbturn or mortise cylinder on interiors.

Auto release, two point flushbolt secures inactive door tops and bottoms followed by normal engagement of retail deadlocks through the lock strike cutout. Retract the deadlock by turn of key and the mechanism releases allowing both door leaves to swing freely.

2.3.7 Provisions for Glazing

Design glazing beads to receive thickness indicated for each glazed assembly. Insulating glass composites for exterior locations have a 1 inch thickness. Glass for interior locations has a 0.25 inch thickness. Coordinate requirements with Section 08 81 00 GLAZING.

PART 3 EXECUTION

3.1 INSTALLATION

Plumb, square, level, and align frames and framing members to receive doors, adjoining side lites, and adjoining window walls. Anchor frames to adjacent construction as indicated and in accordance with manufacturer's printed instructions and the approved shop drawings. Install anchorage that complies with applicable structural requirements. Anchor bottom of each frame to rough floor construction with 3/32 inch thick minimum stainless steel angle clips secured to back of each jamb and to floor construction; use stainless steel bolts and expansion rivets for fastening clip anchors. Hang doors to produce clearances specified in paragraph ALUMINUM DOORS. After erection and glazing, adjust doors and hardware to operate properly.

3.2 Thresholds

Set thresholds in a bed of mastic and secure the thresholds. Protect

aluminum members in contact with masonry, steel, concrete, or dissimilar materials using nylon pads or a bituminous coating. Shim and brace the aluminum system before anchoring the system to the structure. Verify that weep holes are open, and the metal joints are sealed in accordance with the manufacturer's installation instructions. Seal metal-to-metal joints using a sealant recommended by the system manufacturer.

3.3 PROTECTION FROM DISSIMILAR MATERIALS

3.3.1 Dissimilar Metals

Where aluminum surfaces come in contact with metals other than stainless steel, zinc, or small areas of white bronze, protect from direct contact to dissimilar metals.

3.3.1.1 Protection

Provide one of the following systems to protect surfaces in contact with dissimilar metals:

- a. Paint the dissimilar metal with one coat of heavy-bodied bituminous paint.
- b. Apply elastomeric sealant between aluminum and dissimilar metals in accordance with Section 07 92 00 JOINT SEALANTS.
- c. Paint dissimilar metals with one coat of primer and one coat of aluminum paint.
- d. Use a non-absorptive tape or gasket in permanently dry locations.

3.3.2 Drainage from Dissimilar Metals

In locations where drainage from dissimilar metals has direct contact with aluminum, provide protective paint to prevent aluminum discoloration.

3.3.3 Masonry and Concrete

Provide aluminum surfaces in contact with mortar, concrete, or other masonry materials with one coat of heavy-bodied bituminous paint.

3.4 SEALING AROUND ASSEMBLIES

Seal all penetrations of the air barrier by sealing around door openings as necessary to achieve compliance with air leakage requirements indicated in the air barrier sections of the specifications. Flash all doors with corrosion resistant flashing to prevent water intrusion.

3.5 CLEANING

Upon completion of installation, clean door and frame surfaces in accordance with door manufacturer's written recommended procedure. Do not use abrasive, caustic, or acid cleaning agents.

3.6 PROTECTION

Protect doors and frames from damage and from contamination by other materials such as cement mortar. Prior to completion and acceptance of the work, restore damaged doors and frames to original condition, or replace with new ones.

3.7 HARDWARE SCHEDULE

DL = Deadlock; FMC = Full Metal Cover; HD = Heavy Duty; MIL = Mill Finished Aluminum; LHR = Left Hand Reverse; RHR = Right Hand Reverse; SFIC = Small Format Interchangeable Core

Heavy Duty Exterior Openings

Door No. 101A

HWS-1 Exterior DL Entrance Pair

	2 each HD Roton, A31031G	Clear Anodized	
Section 08 71 00	2 each FMC Closer, C02231 with spring stop, stop mounted slide track arm push side mount	630	
	Active Leaf RHR		
	1 each Concealed Retail Deadlock entry by keyed trim, public use while retracted	628	
	1 each Thumbturn	628	
Section 08 71 00	1 each Mortise Cylinder Housing, F25, E09251	630	
Section 08 71 00	1 each 7 Pin SFIC, F24, E09241	630	
	Inactive Leaf LHR		
	1 each Auto Release, Two Point Flush Bolt	626	
Section 08 71 00	1 each Dust Proof Strike, L04021	626	
Section 08 71 00	2 each 4 x 16 Beveled Push Plate, J301	630	
Section 08 71 00	2 each 4 x 16 Beveled Pull Plate, J405	630	
Section 08 71 00	1 each 76" Drip Ledge	MIL	
	1 each 6070 Weatherstrip	Clear Anodized	
	1 each Brush Astragal	Black	
	2 each 36" Bottom Sweep	Clear Anodized	
	1 each 72" Threshold	MIL	
Section 08 71 30	2 each Floor Stop, L02131	626	
GFGI	1 each DPDT Balanced Magnetic Switch	MIL	
GFGI	1 each Electric Strike	630	
GFGI	2 each Card Reader, Wall Mounted	Black	
GFGI	2 each Electric hinge	630	

Note: Furnish electric metallic tubing raceway, cabling, equipment and programming to support connected devices and operate access control components

Note: Furnish electric metallic tubing raceway, pull strings and blank stainless steel coverplates to receive Government furnished and installed devices

Note: Integrate access control equipment with fire alarm equipment. Electric strike fails secure upon fire alarm notification

Note: Access control equipment is located in Comm 117

Heavy Duty Interior Openings

Door No. 101B

HWS-2 DL Foyer Air Lock Pair

	2 each HD Roton, A31031G	Clear Anodized
Section 08 71 00	2 each FMC Closer, C02021	630
Section 08 71 00	2 each 4 x 16 Beveled Push Plate, J301	630
Section 08 71 00	2 each 4 x 16 Beveled Pull Plate, J405	630
	1 each 6070 Weatherstrip	Clear Anodized
	1 each Brush Astragal	Black
	1 each 72" Threshold	MIL
Section 08 71 00	2 each Wall Stop, L02251 or	626
	Floor Stop, L02161	

-- End of Section --